Hospital Readmission Prediction for Diabetic Patients

Using Machine Learning and Explainable AI

Phase 2 Progress Report

Dataset: UCI Diabetes 130-US Hospitals (1999–2008)
Task: 3-class readmission prediction (No Readmit / >30 Days / <30 Days)

Team Members

Name	Student ID	Assigned Work
Bassant Osama	202202308	Data Preprocessing · Logistic Regression · SHAP interpretability
Esraa Magdy	202202198	Exploratory Data Analysis · Extra Trees · SHAP interpretability
Toqa Hassan	202201868	Feature Selection · XGBoost · SHAP interpretability · GitHub organisation
Habib Ahmed	202202178	SVM · LIME & SHAP interpretability

1. Introduction

Hospital readmission within 30 days is a key quality indicator in healthcare and carries financial penalties for institutions. This project builds on two published studies — Rafie et al. (2025) and Dharmarathne et al. (2024) — both of which applied XGBoost with SHAP to diabetes-related prediction tasks. Our work extends them by framing the problem as a three-class classification: no readmission (class 0), readmission after more than 30 days (class 1), and readmission within 30 days (class 2). Class 2 is the most clinically urgent, representing patients who deteriorate quickly after discharge, and is also the hardest to predict because it accounts for only about 11% of all records.

Four models were implemented independently by each team member: XGBoost (Toqa), Logistic Regression (Bassant), Extra Trees (Esraa), and Support Vector Machine (Habib). Each

notebook includes its own interpretability analysis. The shared preprocessing and EDA pipeline is described in Sections 2–5.

2. Dataset

The UCI Diabetes 130-US Hospitals dataset covers clinical records from 130 US hospitals between 1999 and 2008. It includes demographics, admission details, diagnosis codes, medication records, and the readmission outcome.

Attribute	Value
Original records	101,766
After removing invalid race / gender	95,818
After IQR-based outlier removal	84,886
Features before engineering	27 (after initial column drops, it was 50 at beginning)
Features after one-hot encoding	88
Features after Variance Threshold	80
Class 0 — No Readmit	44,772 (52.7%)
Class 1 — Readmitted > 30 days	30,442 (35.9%)
Class 2 — Readmitted < 30 days	9,672 (11.4%)

3. Data Preprocessing

Responsible: Bassant Osama (202202308)

3.1 Missing Data

Two columns had extreme missingness and were dropped rather than imputed: max_glu_serum (~95% missing) and A1Cresult (~83% missing). Imputing at this scale would introduce more bias than the columns remove. Records with invalid race entries ('?') and a gender value of 'Unknown/Invalid' were also excluded, bringing the dataset to 95,818 rows.

3.2 Column Removal

Column(s)	Reason
encounter_id, patient_nbr	Identifiers — carry no predictive signal

Column(s)	Reason
weight	97% missing
max_glu_serum, A1Cresult	>83% missing — dropped rather than imputed
payer_code, medical_specialty	High missingness; low clinical relevance to readmission
17 rare medications	Near-zero variance: effectively all values are 'No'
Combination drug columns	Derived redundantly from base drug columns

3.3 Discharge Disposition Grouping

The 26 discharge codes were collapsed into six clinically coherent groups: Home, Care (skilled nursing or rehab), Transfer, Death, Hospice, and Other. Patients coded as 'Unknown' (code 18) were removed.

3.4 ICD-9 Diagnosis Mapping

Each of the three diagnosis columns contained thousands of raw ICD-9 codes. These were mapped to 17 disease categories using numeric range thresholds and prefix rules for V-codes (Supplementary) and E-codes (External Causes), reducing cardinality while retaining clinical meaning.

3.5 Target Variable Encoding

The readmitted column was recoded as integers: NO → 0, >30 → 1, <30 → 2. The ordering preserves clinical urgency, with class 2 representing the highest-risk outcome.

4. Exploratory Data Analysis

Responsible: Esraa Magdy (202202198)

4.1 Class Distribution

The target is moderately imbalanced: class 0 accounts for 52.7% of records, class 1 for 35.9%, and class 2 for only 11.4%. A classifier that always predicts class 0 would achieve about 53% accuracy without learning anything useful, so this is the minimum baseline all models must exceed.

4.2 Readmission Rates by Age Group

Readmission counts rise steadily from age group [30–40) onward, peaking at [70–80). All three readmission classes reach their highest absolute counts in this bracket. Early readmission (class 2) also grows proportionally with age, consistent with the clinical finding that older patients with multiple comorbidities deteriorate faster after discharge.

4.3 Insulin Usage

Patients whose insulin dose was reduced ('Down') had a higher proportion of no-readmission outcomes relative to those with no insulin or an increased dose. The 'No insulin' group has the largest total count but also the highest absolute number of early readmissions, partly because it is also the largest group.

4.4 Outlier Removal

IQR-based clipping was applied to four features whose outliers likely reflect data entry errors: num_lab_procedures, num_medications, number_diagnoses, and time_in_hospital. This removed 10,932 records, reducing the dataset from 95,818 to 84,886. Utilisation counts (number_outpatient, number_emergency, number_inpatient) were not clipped because extreme values there represent genuine high-utiliser patients.

4.5 Log Transformation

Feature	Skewness before	Skewness after
number_emergency	22.29	3.55
number_outpatient	8.67	2.67
number_inpatient	3.60	1.43
num_medications	1.31	−0.50
time_in_hospital	1.15	0.11
num_lab_procedures	−0.23	−2.19

number_emergency and number_outpatient remain skewed after log1p transformation because most patients have zero prior visits — a zero-inflated pattern that is well-tolerated by tree-based models.

4.6 Feature Distributions

Pre-transformation histograms confirmed strong right skew in number_emergency (22.3) and number_outpatient (8.7), validating the log treatment. After transformation, time_in_hospital reached near-symmetry (skewness 0.11). num_lab_procedures was near-symmetric before transformation; its skewness shifted after log due to a ceiling effect on moderate values.

4.7 Correlation Analysis

- number_inpatient has the strongest linear association with readmitted (r = 0.23), consistent with prior hospitalisation as a key predictor.
- time_in_hospital and num_medications are moderately correlated (r = 0.47) — longer stays involve more complex medication regimens.

- admission_type_id (r = −0.01) and num_procedures (r = −0.04) show negligible association with the target, justifying their removal.
- No feature pair exceeds r = 0.50, so multicollinearity is not a concern for any model.

5. Feature Engineering and Selection

Responsible: Toqa Hassan (202201868)

5.1 Feature Engineering

- Age one-hot encoding: The ordered age bands were expanded into 10 binary columns to allow non-linear age effects without imposing a false ordinal scale.
- Medication ordinal encoding: Seven drug columns were mapped to No=0, Steady=1, Down=2, Up=3, capturing dose direction as a clinically meaningful ordinal variable.
- Categorical one-hot encoding: race, gender, change, diabetesMed, discharge_group, and the three diagnosis columns were expanded using pd.get_dummies (drop_first=True).
- admission_source_id encoding: Integer source codes were cast to strings and one-hot encoded; codes appearing in very few records were dropped.

5.2 Filter-Based Feature Selection

- Correlation-based removal: admission_type_id and num_procedures were dropped due to near-zero correlation with the target (r = −0.01 and −0.04).
- Sparse column drop: Dummy columns with fewer than 300 positive values across 84,886 rows (< 0.35% prevalence) were removed. Eight columns were dropped, including age_[0–10) with only 98 patients.
- Variance Threshold: sklearn's VarianceThreshold (threshold = 0.0099) removed a further eight near-constant features, including age_[10–20), race_Asian, discharge_group_Hospice, and three sparse diagnosis flags.

Mutual Information SelectKBest was tested at 90% and 98% cumulative thresholds but reduced XGBoost test accuracy by up to 9 points. XGBoost handles selection internally via tree splits, so aggressive external filtering was counterproductive. The final feature matrix contains 80 features.

6. Model 1 — XGBoost

Responsible: Toqa Hassan (202201868)

6.1 Why XGBoost

XGBoost directly replicates the best-performing model in both reference papers and is well-suited to structured clinical data. It builds an ensemble of trees where each corrects the residuals of the previous one, handles mixed numerical and binary inputs natively, supports multi-class softmax probabilities, and integrates with SHAP's exact TreeExplainer.

6.2 Hyperparameter Tuning

Initial parameters were found by RandomizedSearchCV (30 iterations, 5-fold CV). The learning curve from that run showed a bias-variance gap of 0.059 — moderate overfitting. Parameters were then manually regularized: max_depth reduced from 6 to 4, min_child_weight raised from 1 to 3, colsample_bytree reduced from 1.0 to 0.8, and L1/L2 penalties added.

Parameter	Initial best	Final (regularized)
n_estimators	200	200
max_depth	6	4
learning_rate	0.1	0.1
subsample	0.6	0.6
colsample_bytree	1.0	0.8
min_child_weight	1	3
gamma	0.2	0.3
reg_alpha (L1)	—	0.1
reg_lambda (L2)	—	2.0

6.3 Training Setup

The 80-feature matrix was split 80/20 with stratification. SMOTE was tested but increased noise near the three-class boundary and reduced accuracy below the baseline; it was dropped. Seven continuous features were standardised with StandardScaler after the split to prevent leakage.

6.4 Results

Phase	Accuracy	Precision (wtd)	Recall (wtd)	F1 (wtd)	AUC (OvR)
Training	59.0%	58.3%	59.0%	54.4%	69.9%
Testing	57.8%	56.0%	57.8%	52.9%	66.9%

The training-to-test accuracy gap is 1.2 points, down from 4.1 before regularization. The bias-variance gap on the learning curve narrowed to 0.017. The model outperforms the naive majority-class baseline of 52.7%.

Class	Precision	Recall	F1	Support
No Readmit (0)	—	84.2%	—	8,978
>30 Days (1)	—	36.3%	—	6,088
<30 Days (2)	—	2.9%	—	1,935

Class 2 recall of 2.9% is the main limitation — a known consequence of not using resampling on a three-class imbalanced problem.

6.5 Learning and Validation Curves

The learning curve shows CV accuracy flattening around 25,000 samples and stabilising at ~57.8%. The performance ceiling is set by the available features, not data volume. The validation curve confirms CV accuracy peaks at depth 4–6; training accuracy rises sharply beyond depth 6, marking the overfitting zone. Depth 4 was chosen as the most conservative point within the peak.

6.6 SHAP Interpretability

- number_inpatient is the single most important feature across all three classes. High prior inpatient visits strongly increase predicted risk of early readmission.
- discharge_group_Death shows very large positive SHAP for class 0 and large negative SHAP for classes 1 and 2 — patients discharged to death cannot be readmitted, which the model captures precisely.
- number_emergency increases readmission risk across all classes.
- number_outpatient high values push toward class 1 (>30 days), consistent with managed chronic-disease patients who return on a planned basis.

Local waterfall plots for four test patients show predictions driven mainly by utilisation history. Instance 4, predicted as >30 days, is driven by high number_outpatient (+0.21) and prior inpatient admissions (+0.17).

7. Model 2 — Logistic Regression

Responsible: Bassant Osama (202202308)

7.1 Why Logistic Regression

Logistic Regression was chosen as the interpretable linear baseline, following the approach of Dharmarathne et al. (2024). It produces directly readable coefficients without any post-hoc tool, satisfies well-understood statistical assumptions, and serves as a meaningful reference point: if a linear and non-linear model perform similarly, the simpler one is preferred.

7.2 Pipeline and Configuration

The model was wrapped in an imblearn Pipeline (SMOTE → StandardScaler → LogisticRegression) so that oversampling is applied only inside each training fold, preventing data leakage.

Step	Component	Settings
1	SMOTE	random_state=42 — minority class upsampling within folds only
2	StandardScaler	Zero mean, unit variance — essential for coefficient comparability
3	LogisticRegression	solver=lbfgs, max_iter=1000, C=1.0, class_weight=balanced

class_weight=balanced was set because the 11% minority class would otherwise be almost entirely ignored. C=1.0 applies default L2 regularization.

7.3 Assumption Checks

- Multicollinearity: No feature pair exceeds r = 0.50 in the correlation matrix, so coefficients are stable.
- Feature scaling: StandardScaler applied — all coefficients are directly comparable.
- Independence: Each row is a unique encounter. The same patient can appear across multiple years, which is a known limitation that would require patient-level clustering to resolve properly.

7.4 Cross-Validation Results (10-Fold Stratified)

Metric	CV mean	95% CI	Note
Accuracy	49.62%	± 1.33%	Stable across folds — no overfitting
F1 weighted	49.15%	± 1.23%	Consistent
AUC-ROC (OvR weighted)	61.12%	± 1.53%	Above 50% random baseline

Narrow confidence intervals confirm no overfitting. AUC of 61.12% shows genuine discrimination ability despite raw accuracy sitting near the majority-class baseline.

7.5 Hold-out Test Set Performance

Metric	Score
Accuracy	50.17%
F1-Score (weighted)	49.85%

Metric	Score
AUC-ROC (OvR weighted)	61.79%
Average (Acc + F1 + AUC) / 3	53.94%

Class	Precision	Recall	F1	Support
No Readmit (0)	0.62	0.69	0.65	9,935
>30 Days (1)	0.48	0.29	0.36	6,697
<30 Days (2)	0.17	0.29	0.21	2,108
Weighted avg	0.52	0.50	0.50	18,740

Class 2 recall of 29% is substantially higher than XGBoost's 2.9%. class_weight=balanced and SMOTE together force the linear boundary to allocate more space to the minority class, at the cost of overall accuracy. For clinical triage where missing a high-risk patient is costly, this tradeoff is clinically meaningful.

7.6 Confusion Matrix

	Pred: No Readmit	Pred: >30 Days	Pred: <30 Days	Row Recall
Actual: No Readmit (0)	6,878	1,451	1,606	69.2%
Actual: >30 Days (1)	3,333	1,910	1,454	28.5%
Actual: <30 Days (2)	908	586	614	29.1%

The largest off-diagonal block is the 3,333 class 1 patients predicted as class 0 — the linear boundary cannot cleanly separate late readmission from no readmission. Class 2 predictions are spread across all three columns, confirming the model's limited resolution for the minority group. ROC AUC per class: class 0 = 0.627, class 1 = 0.610, class 2 = 0.596. The three curves cluster closely, characteristic of a linear model on a non-linearly separable three-class problem.

7.7 SHAP Interpretability

SHAP LinearExplainer was applied to the standardised test matrix. The all-class bar plot shows age group dummies dominating: the top eight features by mean |SHAP| are all age categories, with age_[70–80) reaching ~3.7. Age dummies carry the largest model coefficients (up to 1.34 mean |coef|) and create the strongest linear separations between classes. The first non-age feature is number_inpatient at rank 8.

For class 2 specifically, all age groups have negative SHAP values — older age is associated with class 0 or class 1 in the LR model because the linear boundary cannot capture the non-linear age-by-severity interaction. The only feature with a clear positive SHAP for class 2 is number_inpatient (up to +1.0), confirming prior hospitalisation as the primary early-readmission signal even under a linear model.

Three local waterfall plots were generated for class 2. All three show $f(x) \approx E[f(X)] = -0.305$, meaning the displayed features (num_medications, time_in_hospital, num_lab_procedures) contribute almost nothing to the prediction — they are the three lowest-coefficient features. More informative local explanations would require selecting patients where number_inpatient or age varies substantially.

7.8 Coefficient Analysis

- 21 features KEEP (|coef| ≥ 0.05): All age groups (0.37–1.34), number_inpatient (0.21), all seven medications (0.05–0.09), number_emergency (0.07), num_medications (0.06), insulin (0.05), number_diagnoses (0.05).
- 3 features REMOVE (|coef| < 0.05): number_outpatient (0.047), time_in_hospital (0.036), num_lab_procedures (0.016).

A feature ablation test showed that dropping all three improved class 2 AUC by +0.57% but reduced overall accuracy by 3.4 points. The final decision was to retain all 24 features — the marginal AUC gain does not justify the accuracy loss.

8. Model 3 — Extra Trees

Responsible: Esraa Magdy (202202198)

8.1 Why Extra Trees

The Extra Trees Classifier (Extremely Randomised Trees) was chosen for its stronger variance reduction compared with standard Random Forest. Where Random Forest selects the best split from a random feature subset, Extra Trees selects both the feature and the split threshold randomly, adding an extra layer of regularisation. This makes it particularly effective at suppressing the statistical noise present in the high-dimensional, imbalanced clinical dataset, and it natively supports multi-class classification with class-weight adjustment.

8.2 Preprocessing Steps

- Variance Threshold (0.01): 23 redundant low-variance features were automatically removed, reducing the feature space from 83 to 60 columns and improving computational efficiency by approximately 30%.
- Recursive Feature Elimination (RFE): The top 10 most clinically influential features were identified, allowing the model to focus on genuine signals rather than noise.
- StandardScaler: Applied to all continuous features for numerical stability.

- Class imbalance: Handled via class_weight='balanced' within the algorithm itself, which penalises misclassifications of the minority class without introducing synthetic noise from SMOTE.

8.3 Hyperparameter Tuning

Initial models with max_depth=20 achieved ~98% training accuracy while validation accuracy stalled at ~52% — a clear memorisation problem. Systematic validation curve analysis identified max_depth=12 as the point where CV accuracy plateaus before the training-validation gap widens further.

Parameter	Initial value	Final (regularized)
n_estimators	500	1,000
max_depth	20	12
min_samples_leaf	1	10
class_weight	balanced	balanced
max_features	sqrt	sqrt

Raising min_samples_leaf from 1 to 10 was the single most effective change: it prevented individual trees from fitting to very small patient subgroups, which was the main source of the original 40%+ training-validation gap.

8.4 Results

Phase	Accuracy	Precision (wtd)	Recall (wtd)	F1 (wtd)
Training	53.37%	—	—	—
Testing	48.42%	54.0%	47.0%	49.0%

The training-to-test gap is 4.95 points, reduced from more than 40 points in the initial unregularised model. The model generalises to unseen data at a level comparable with the other models in this study.

Class	Precision	Recall	F1	Support
No Readmit (0)	66%	59%	62%	15,338
>30 Days (1)	46%	28%	35%	10,194
<30 Days (2)	18%	51%	27%	3,214

Class 2 recall of 51% is the highest across all four models. Identifying more than half of the patients who will be readmitted within 30 days is a clinically significant result — these are the patients for whom early intervention (closer follow-up, medication review, targeted discharge planning) can prevent deterioration. The 18% precision means roughly one in five class 2 flags is correct, which is acceptable for a screening tool where false negatives are more costly than false positives.

8.5 Learning and Validation Curves

The learning curve shows the training and cross-validation lines converging and stabilising at approximately 40,000 samples, confirming the model has learned the underlying data patterns and is no longer overfitting. The validation curve for max_depth confirms that accuracy peaks at depth 12; beyond that, training accuracy continues rising while validation accuracy stagnates, marking the start of the overfitting regime.

8.6 SHAP Interpretability

Feature importance from the trained model highlights three clinically coherent drivers:

- number_inpatient: The strongest predictor. Patients with multiple prior hospitalisations are at significantly higher risk of being readmitted within 30 days.
- num_medications and time_in_hospital: Together they represent clinical intensity — complex cases with longer stays and more medications have a higher likelihood of return.
- discharge_group_Death: Correctly carries near-zero feature importance for readmission classes 1 and 2, demonstrating logical consistency: a patient discharged to death cannot be readmitted.

Full SHAP beeswarm and dependence plots are in development for the next phase, alongside LIME local explanations for individual high-risk patient cases.

9. Model 4 — Support Vector Machine

Responsible: Habib Ahmed (202202178)

9.1 Why SVM

Support Vector Machine with an RBF kernel was implemented to evaluate a margin-based classifier that can handle non-linear class boundaries. SVM finds the hyperplane that maximises the margin between classes, and the RBF kernel maps inputs into a higher-dimensional space where the classes may become more separable.

9.2 Training Setup

The dataset was split 80/20 with stratification to preserve the class distribution across both subsets. The RBF kernel was selected for its ability to model complex feature interactions in the high-dimensional encoded feature space.

9.3 Results

Metric	Score
Overall Accuracy	69%
Class 0 — No Readmit	Good performance
Class 1 — >30 Days	Lower performance
Class 2 — <30 Days	Weakest — most important clinically

SVM achieves the highest overall accuracy of the four models (69%). Class 2 performance is weak, which is expected without explicit class-weight balancing or resampling.

9.4 Hyperparameter Tuning

GridSearchCV was used to tune C (regularisation strength) and gamma (RBF kernel width). The best parameter combination improved model performance over the default configuration.

9.5 Interpretability

- LIME (Local Interpretable Model-agnostic Explanations): Applied at the instance level. For each patient prediction, LIME fits a local linear approximation and highlights which features pushed the prediction toward a particular class. This makes it possible to explain individual clinical decisions to healthcare staff.
- SHAP: Applied for global feature importance across the test set. Number of medications and length of hospital stay emerged as the strongest predictors. Certain diagnosis categories also showed notable influence on the predicted readmission class.

10. Cross-Model Comparison

SVM achieves the highest overall accuracy (69%), though class 2 performance is weak. Extra Trees reaches the best class 2 recall (51%), making it the strongest choice when the clinical goal is to identify high-risk patients early. XGBoost has the best-documented AUC and the most thorough regularisation analysis. Logistic Regression is the most interpretable model with directly readable coefficients and the second-best class 2 recall (29%).

No single model is best across all metrics. In practice, a two-stage approach could work well: use Extra Trees or Logistic Regression as a screening layer to flag class 2 candidates, then apply XGBoost for overall risk stratification. This trades some precision for substantially better recall on the most critical patient group.

11. Challenges

Challenge	Model(s) affected	How it was handled
Multi-class imbalance (~11% class 2)	All	class_weight=balanced (LR, ET); regularization (XGB); no SMOTE in XGB pipeline
SMOTE degraded accuracy in 3-class setting	XGB	Removed; retained inside CV folds for LR only
XGBoost rejected feature names with '[', ']', '<'	XGB	Column names sanitised before training
SHAP 3D indexing for multi-class XGBoost	XGB	beeswarm: shap_values[:,:,cls]; waterfall: shap_values[i,:,cls]
LR waterfall plots showed near-zero contributions	LR	Three lowest-coefficient features were selected — noted as a limitation
Extra Trees overfitting at max_depth=20 (98% train accuracy)	ET	Validation curve identified optimal depth of 12; min_samples_leaf raised to 10
Missing HbA1c (~83%) and glucose (~95%)	All	Unavoidable — represents the primary accuracy ceiling for this dataset
MI-based selection reduced XGB accuracy by 9 points	XGB	MI retained for documentation only; Variance Threshold used for actual selection

14. References

[1] Rafie, Z. et al. (2025). Leveraging XGBoost and explainable AI for accurate prediction of type 2 diabetes. *BMC Public Health, 25*, 3688. https://doi.org/10.1186/s12889-025-24953-w

[2] Dharmarathne, G. et al. (2024). A novel machine learning approach for diagnosing diabetes with a self-explainable interface. *Healthcare Analytics, 5*, 100301. https://doi.org/10.1016/j.health.2024.100301

[3] Strack, B. et al. (2014). Impact of HbA1c measurement on hospital readmission rates. *BioMed Research International, 2014*.

[4] Emi-Johnson, O. G. & Nkrumah, K. J. (2025). Predicting 30-day hospital readmission in patients with diabetes using machine learning on electronic health record data. *Cureus, 17*(4), e82437. https://doi.org/10.7759/cureus.82437

[5] Lundberg, S. M. & Lee, S. I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*.